Numerical Study of the accuracy of Fermi's golden rule in a spherically symmetric well

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Abstract

In this study, the decay rates of the quasi-bound states of a spherical well were computed. The parameters varied were the well's radius, potential and the angular quantum number. The decay rates were computed using Fermi's golden rule. These decay rates were then compared with decay rates computed using partial wave analysis. It was found that decay rates increase with decreasing values of radius and potential, and with increasing values of angular quantum number. This relation corresponds with the well's width, as decay rates increase when the well becomes narrower. It was also found that the approximations using Fermi's golden rule are only reasonably accurate for a well which can support a large range of energies in its quasi-bound state. This occurs for small values of angular quantum number and radius, as well as a large potential depth. From our results, we found that a value of radius $R \approx 1$, angular quantum number $l \approx 1$ and potential $U \geq 20$ will result in a low percentage error for decay rates approximated by Fermi's golden rule.

1 Introduction

1.1 Background Theory

1.1.1 Resonances

Bound states are states which are localized to a region of space. They occur in systems which consist of potential wells. The simplest model is a 1D finite square well, which is described by the potential $V_0(x) = -U\Theta(a - |x|)$, where $\Theta(x)$ is the unit step function, $a > 0$ and $U > 0$.

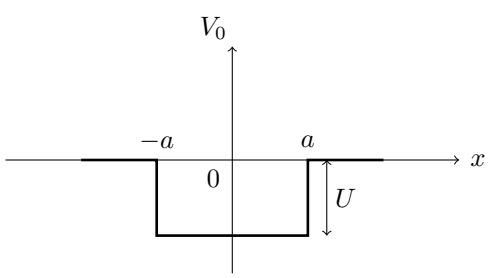

Figure 1: 1D Square Well

Note that the bound states occur for negative energies E which are in the range $-U \leq E \leq 0$. States with energy $E > 0$ are known as free states. This is because, for these energies, the particles would be delocalized in space. Now we add a deconfining potential $V_1(x) = V_b\Theta(b - |x|)$, where $b > a$ and $V_b > U$, to the the square well such that the total potential is $V(x) = V_0(x) + V_1(x)$.

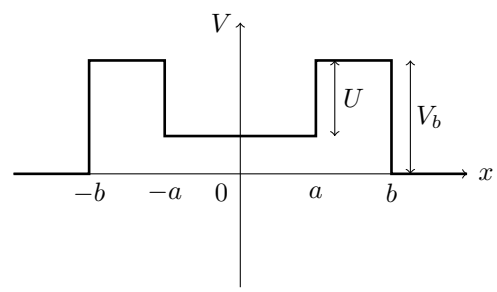

Figure 2: 1D Square Well with deconfining potential

Note that now the "bound states" in figure 2 exist at the positive energy range $(V_b - U) \leq E \leq V_b$. These shifted bound states are known as the quasi-bound states. Unlike the true bound states, they are only approximate eigenstates of the Hamiltonian $\hat{H} = \hat{T} + \hat{V}$. This means that these quasi-bound states would decay under time evolution.

Now let us treat the potential $V(x)$ as a scatterer. If we were to conduct a scattering experiment in which plane waves with a range of energies E are incident on $V(x)$, we would find that at certain energies, there would be a sharp increase in the scattering cross section σ. If we were to plot $|\psi(x)|^2$ where $\psi(x)$ is the total wavefunction at these energies, we would observe that $|\psi(x)|^2$ is much larger within the potential region $-b \leq x \leq b$. The enhancement of $|\psi(x)|^2$ and σ at these energies is known as a scattering resonance. Thus, these energies are called resonance energies E_{res}.

A particle incident at $E = E_{\text{res}}$ would be trapped in the region $-a \leq x \leq a$ temporarily. However, it would still be able to escape via the quantum tunneling effect. This corresponds to the decay of a quasi-bound state. In other words, when a scattering resonance occurs, the particle would be trapped in a quasi-bound state for some finite lifetime. The decay rate of quasi-bound states is related to the full width at half maximum (FWHM) of the scattering cross section σ.

1.1.2 Fermi's golden rule

The decay rate κ of the quasi-bound state can be approximated using Fermi's golden rule,

$$\kappa = \frac{2\pi}{\hbar}\left|\langle\psi_{k(E_{\text{res}})}|\hat{V}_1|\varphi\rangle\right|^2 \mathcal{D}(E_{\text{res}}) \tag{1}$$

where $\langle\psi_k(E_{\text{res}})|$ are free particle states with energy $E = E_{\text{res}}$, $\hat{V}_1$ is our deconfining potential and $\mathcal{D}(E_{\text{res}})$ is the density of states evaluated at $E = E_{\text{res}}$. $|\varphi\rangle$ are the eigenstates of the hamiltonian $\hat{H}_0 = \hat{T} + \hat{V}_0$.

$$\hat{H}_0|\varphi\rangle = E|\varphi\rangle \tag{2}$$

This means that $|\varphi\rangle$ are the bound states of the confining potential V_0. The matrix element $\langle\psi_{k(E_{\text{res}})}|\hat{V}_1|\varphi\rangle$ can be interpreted as the particle starting in quasi-bound state $|\varphi\rangle$ and going through a transition to the free states $|\psi_{k(E_{\text{res}})}\rangle$ due to the deconfining potential $\hat{V}_1$. The matrix element is therefore known as the transition amplitude.

1.1.3 Spherical Well

The system being studied in this project is a spherical well with the following confining potential,

$$V_0(r) = \begin{cases} -U & \text{for} \quad r \leq R \\ 0 & \text{otherwise} \end{cases} \tag{3}$$

where $U > 0$. Such a potential is known as a central potential as it only depends on the radial coordinate r. In this case, the solution to the time-independent Schrodinger equation would take the following separable form,

$$\Psi(r, \theta, \phi) = R_l(r)Y_{lm}(\theta, \phi) \tag{4}$$

where $Y_{lm}(\theta, \phi)$ are the spherical harmonics and $A_l(r)$ are solutions to the following radial equation.

$$\left[-\frac{\hbar^2}{2m}\left(\frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr}\right) + \frac{l(l+1)\hbar^2}{2mr^2} + V_0(r)\right]R_l(r) = ER_l(r) \tag{5}$$

We can define a new radial function $u_l(r) = rR_l(r)$ and substitute it into equation (5) in order to transform it into the one-dimensional time-independent schrodinger equation.

$$-\frac{\hbar^2}{2m}\frac{d^2u_l(r)}{dr^2} + V(r)u_l(r) = Eu_l(r) \tag{6}$$

Note that the effective potential is $V(r) = V_0(r) + V_1(r)$ where,

$$V_1(r) = \frac{l(l+1)\hbar^2}{2mr^2} \tag{7}$$

From this, we see that the system is being subjected to both a confining (negative) potential $V_0(r)$ and a deconfining (positive) potential $V_1(r)$. This is analogous to the system described in section 1.1.1. A plot of $V(r)$ against r is shown in figure 3. *Note that the definitions for $V_0(r)$ and $V_1(r)$ will follow that of equation (3) and (7) for the remainder of this report.

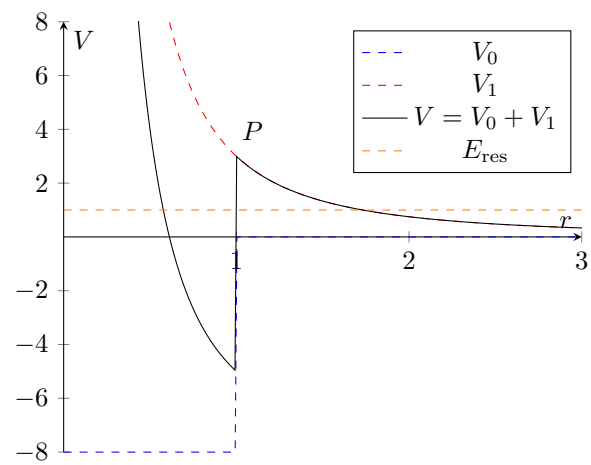

Figure 3: Effective potential $V(r)$ of a spherical well

Referring to figure 3, we see that the spherical well can support quasi-bound states for energies near $E \approx E_{\text{res}}$, where $E_{\text{res}} > 0$ but is less than the local maxima at point P. This is similar to the scenario depicted in figure 2. If the bottom of the well in figure 3 were above the horizontal axis, then the quasi-bound states would exist for energies between the bottom of the well and the local maxima at point P.

1.1.4 Partial Wave Analysis

Partial Wave Analysis is a method for analyzing scattering problems involving central potentials. Assume that the incident plane waves are propagating in the z-direction. As the potential is symmetrical about the z-axis, the scattering amplitude f would have no dependence on the azimuthal angle ϕ. We can therefore expand the scattering amplitude f and the total wavefunction Ψ using the Legendre polynomials $P_l(\cos\theta)$ as our basis functions. This is because the Legendre polynomials are complete on the interval $0 \leq \theta \leq \pi$.

$$\Psi_{\vec{k}}(r, \theta) = \sum_{l=0}^{\infty} R_l(k, r)P_l(\cos\theta) \tag{8}$$

$$f(k,\theta) = \sum_{l=0}^{\infty} f_l(k) P_l(\cos\theta) \tag{9}$$

$\vec{k}$ is the wavevector of an incident plane wave. The radial functions $R_l(k,r)$ are the solutions to equation (5) and are known as partial waves. The functions $f_l(k)$ are known as partial amplitudes. Both functions are only dependent on the angular quantum number l and not on the magnetic quantum number m as m does not appear in the radial equation (5). It should be noted that partial wave analysis is only a semi-exact method as it relies on the asymptotic (i.e. when $r \to \infty$) forms of the spherical Bessel and Hankel functions.

1.2 Aim

The objective of this study is to analyze the accuracy of Fermi's golden rule at approximating the decay rates of scattering resonances of our spherical well. This will be done by comparing decay rates obtained from Fermi's golden rule against decay rates obtained from performing partial wave analysis on the spherical well.

2 Methodology

In this section we present our methodology for calculating the decay rate κ using both partial wave analysis and Fermi's golden rule. Derivation of relevant equations will be presented first, followed by the implementation of the numerical calculation.

2.1 Input Parameters

For our spherical well, there were three parameters that we varied: the well depth U, the angular momentum quantum number l and the radius of the well R. Three sets of calculations were carried out, one for each parameter. The numerical calculations produced plots of κ against each parameter. The inputs for each calculation are listed below.

- For the calculation which varied l, $U = 8$ and $R = 1$.
- For the calculation which varied U, $R = 1$ and $l = 1$
- For the calculation which varied R, $U = 20$ and $l = 1$

Also note that for all scattering calculations, incident energy E was varied from 0 to 1000.

2.2 Decay Rate - Partial Wave Analysis

2.2.1 Derivation

Note that the derivation in this section was taken from chapter 13.3 of Quantum Mechanics by Bransden and Joachain.

In this section we will use partial wave analysis to give us a more detailed description of scattering resonances in spherically symmetric scattering problems. The equations derived in this section will be used to calculate the decay rates κ. The radial equation (5) can be rearranged into the spherical Bessel equation,

$$\left[\frac{d^2}{dr^2} + \frac{2}{r}\frac{d}{dr} - \frac{l(l+1)}{r^2} - U(r) + k^2\right] R_l(k,r) = 0 \tag{10}$$

where $U(r) = \frac{2m}{\hbar^2}V(r)$ and $k^2 = \frac{2mE}{\hbar^2}$. For a region $r > d$ where the potential $U(r)$ can be neglected, the solution to the Bessel equation are as follows.

$$R_l(k,r) = B_l(k) j_l(kr) + C_l(k) n_l(kr) \tag{11}$$

where $j_l(kr)$ and $n_l(kr)$ are spherical Bessel functions of the first and second kind respectively. The asymptotic form of (11) as $r \to \infty$ is shown below.

$$R_l(k,r) \to \frac{1}{kr}\left[B_l(k)\sin\left(kr - \frac{l\pi}{2}\right) - C_l(k)\cos\left(kr - \frac{l\pi}{2}\right)\right] \tag{12}$$

We can rewrite the above expression as,

$$R_l(k,r) \to A_l(k)\frac{1}{kr}\sin\left(kr - \frac{l\pi}{2} + \delta_l(k)\right) \tag{13}$$

where $A_l(k) = \sqrt{B_l^2(k) + C_l^2(k)}$ and $\delta_l(k) = -\arctan(C_l(k)/B_l(k))$. The quantity $\delta_l(k)$ is known as a phase shift and it characterizes the strength of the scattering of the lth partial wave. For our spherical well, the phase shift $\delta_l(k)$ takes the following form.

$$\delta_l(k) = \frac{\pi}{2} + \arg\left[k h_l^-(kR)' j_l(qR) - q h_l^+(kR) j_l'(qR)\right] \tag{14}$$

Where $h^+(\rho)$ and $h^-(\rho)$ are spherical Hankel functions of the first and second kind respectively and the wavevectors q and k have the following definitions.

$$q = \sqrt{2m(E+U)/\hbar^2} \tag{15}$$

$$k = \sqrt{2mE/\hbar^2} \tag{16}$$

We will now evaluate the partial cross sections $f_l(k)$. The asymptotic form of total wavefunction $\Psi_{\vec{k}}$ as $r \to \infty$ is shown below.

$$\Psi_{\vec{k}} \to e^{i\vec{k}\cdot\vec{r}} + \frac{f(k,\theta)e^{ikr}}{r} \tag{17}$$

Noting that the wavevector of the incident wave $\vec{k}$ is pointing in the z direction, we can expand the exponential $e^{i\vec{k}\cdot\vec{r}}$ as follows.

$$e^{i\vec{k}\cdot\vec{r}} = \sum_{l=0}^{\infty}(2l+1)i^l j_l(kr) P_l(\cos\theta) \tag{18}$$

The asymptotic form of the above expansion as $r \to \infty$ is as follows.

$$e^{i\vec{k}\cdot\vec{r}} \to \sum_{l=0}^{\infty}(2l+1)(2l+1)i^l(kr)^{-1}\sin(kr - \frac{l\pi}{2})P_l(\cos\theta) \tag{19}$$

Substitute (19) and (9) into (7).

$$\Psi_{\vec{k}} \to \sum_{l=0}^{\infty}\left[(2l+1)(2l+1)i^l(kr)^{-1}\sin(kr - \frac{l\pi}{2}) + f_l(k)\frac{e^{ikr}}{r}\right]P_l(\cos\theta) \tag{20}$$

Comparing (20) with (8), we obtain the following asymptotic form for $R_l(k,r)$.

$$R_l(k,r) \to (2l+1)i^l(kr)^{-1}\sin(kr - \frac{l\pi}{2}) + f_l(k)\frac{e^{ikr}}{r} \tag{21}$$

We can rewrite (13) and (21) using $\sin(x) = (2i)^{-1}(e^{ix} - e^{-ix})$ and compare their coefficients in order to obtain $A_l(k)$ and $f_l(k)$.

$$A_l(k) = (2l+1)i^l e^{i\delta_l(k)} \tag{22}$$

$$f_l(k) = \frac{2l+1}{2ik}[\exp[2i\delta_l(k)] - 1] \tag{23}$$

Substitute (23) into (9).

$$f(k,\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)[\exp(2i\delta_l(k)) - 1]P_l(\cos\theta) \tag{24}$$

From (24) we can calculate the total cross section σ.

$$\sigma = \int |f(k,\theta)|^2 d\Omega = 2\pi \int_{-1}^{1} d(\cos\theta)|f(k,\theta)|^2 \tag{25}$$

$$\sigma = 2\pi \sum_{ll'} f_{l'}^* f_l \int_{-1}^{+1} d(\cos\theta)P_{l'}(\cos\theta)P_l(\cos\theta) \tag{26}$$

Using $\int_{-1}^{+1} dx P_{l'}(x)P_l(x) = \frac{2}{2l+1}\delta_{ll'}$,

$$\sigma = \sum_{l=0}^{\infty}\frac{4\pi}{2l+1}|f_l(k)|^2 = \sum_{l=0}^{\infty}\sigma_l \tag{27}$$

Where the partial cross section σ_l is given by the following expression.

$$\sigma_l = \frac{4\pi}{2l+1}|f_l(k)|^2 \tag{28}$$

We can substitute (23) into (28).

$$\sigma_l = \frac{4\pi(2l+1)}{k^2}\frac{1}{1+\cot^2(\delta_l(k))} \tag{29}$$

We see from (29), that at certain values of $\delta_l(k)$, the partial cross section σ_l would reach a maximum value of $\frac{4\pi(2l+1)}{k^2}$. This corresponds to a scattering resonance. Near a resonance peak, $E \approx E_{\text{res}}$, σ_l would take the form of a Lorentzian curve.

$$\sigma_l(E = E_{\text{res}}) = \frac{4\pi(2l+1)}{k^2}\frac{\kappa^2/4}{(E-E_{\text{res}})^2 + \kappa^2/4} \tag{30}$$

Note that κ is the full width at half maximum (FWHM). It is related to the lifetime τ by $\kappa \approx 1/\tau$. It can therefore be treated as the decay rate of a quasi-bound state.

2.2.2 Implementation

Partial wave analysis calculation of κ was implemented numerically in Python using the following procedure.

1. The phase shift $\delta_l(k)$ was obtained using equation (14).
2. The partial scattering amplitude $f_l(k)$ was obtained using equation (23).
3. The partial cross section σ_l was obtained using using equation (28).
4. κ was obtained by plotting σ_l against E and obtaining the FWHM from the plot. This is illustrated in figure 4.

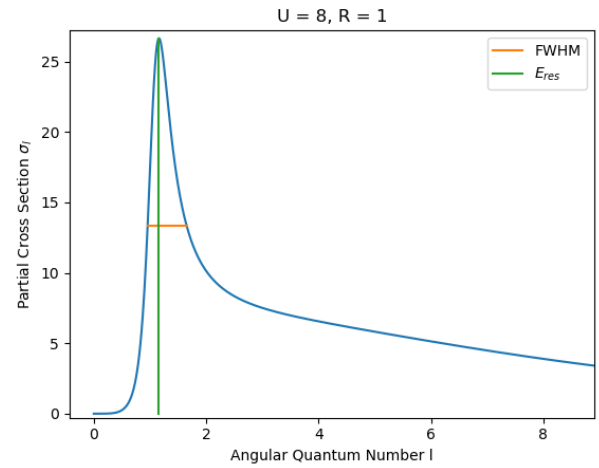

Figure 4: σ_l peaks against l

The above procedure was carried out for each of the parameters listed in section 2.1.

2.3 Decay Rate - Fermi's golden rule

2.3.1 Derivation

Here we will derive an expression for the decay constant κ using Fermi's golden rule. Upon solving equation (5) for our spherical well, we obtain the following solution for $R_l(r)$.

$$R_l(r) = \begin{cases} \alpha_l j_l(qr) & r \le R \\ \beta_l[j_l(kr) - \tan(\delta_l(k))n_l(kr)] & r \ge R \end{cases} \tag{31}$$

Where α_l and β_l are normalization constants and $\tan(\delta_l(k))$ has the following form.

$$\tan(\delta_l(k)) = \frac{kj_l'(kR)j_l(qR) - qj_l(kR)j_l'(qR)}{ky_l'(kR)j_l(qR) - qy_l(kR)j_l'(qR)} \tag{32}$$

Referring to equation (7), we note that the deconfining potential $V_1(r) = 0$ when $l = 0$. Hence, the bound states $\varphi(\vec{r})$ are obtained from (31) by setting $l = 0$. Note that $m = 0$ since $-l \le m \le l$.

$$\varphi(\vec{r}) = \Psi_{00}(\vec{r}) = R_0(r)Y_{00}(\theta,\phi) \tag{33}$$

The states $\psi_k(E_{\text{res}})$ are approximated as free particle states, i.e. states which would exist as if $V(r) = 0$.

$$\psi_k(E_{\text{res}},\vec{r}) = \frac{1}{(2\pi)^{3/2}}e^{i\vec{k}\cdot\vec{r}} \tag{34}$$

For simplicity, we will assume that the plane wave state $\psi_k(E_{\text{res}},\vec{r})$ is propagating the z-direction so that $\vec{k}\cdot\vec{r} = kr\cos\theta$. We will now proceed to calculate the matrix ele-

ment $\langle\psi_k(E_{\text{res}})|\hat{V}_1|\varphi\rangle$.

$$\begin{aligned}
\langle\psi_k(E_{\text{res}})|\hat{V}_1|\varphi\rangle &= \int d^3r\langle\psi_k(E_{\text{res}})|\vec{r}\rangle\langle\vec{r}|\hat{V}_1|\varphi\rangle \\
&= \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta d\theta \quad \times \\
&\quad \int_0^{\infty} drr^2\psi_k(E_{\text{res}},\vec{r})V_1(r)R_0(r)Y_{00}(\theta,\phi) \\
&= (2\pi)\int_0^{\pi}\sin\theta d\theta\int_0^{\infty} drr^2\frac{e^{-i\vec{k}\cdot\vec{r}}}{(2\pi)^{3/2}}\frac{l(l+1)\hbar^2}{2mr^2}\frac{R_0(r)}{\sqrt{4\pi}} \\
&= \frac{\pi[l(l+1)]\hbar^2}{(2\pi)^{3/2}m\sqrt{4\pi}}\int_0^{\infty} drR_0(r)\int_0^{\pi} d\theta\sin\theta e^{-ikr\cos\theta} \\
&= \frac{l(l+1)\hbar^2}{2m\sqrt{8\pi^2}}\int_0^{\infty} drR_0(r)\int_{-1}^{1} due^{ikru} \\
&= \frac{[l(l+1)]\hbar^2}{m\sqrt{8\pi^2}}\int_0^{\infty} dr\frac{\sin kr}{kr}R_0(r) \\
&= \frac{l(l+1)\hbar^2}{m\sqrt{8\pi^2}}\Bigg[\alpha_0\int_0^{R} j_0(kr)j_0(qr) \quad + \\
&\quad \beta_0\int_R^{\infty} j_0(kr)[j_0(kr)-\tan(\delta_0(k))y_0(kr)]\Bigg] \\
&= \frac{l(l+1)\hbar^2}{2\pi m\sqrt{2}}[\mathcal{I}_A(q,k,R)+\mathcal{I}_B(q,k,R)] \\
&= \frac{l(l+1)\hbar^2}{2\pi m\sqrt{2}}[\mathcal{I}_A(q,k,R)+\mathcal{I}_B(q,k,R)] \qquad (35)
\end{aligned}$$

Where we have defined two integrals,

$$\mathcal{I}_A(q,k,R) = \alpha_0\int_0^{R} drj_0(qr)j_0(kr) \qquad (36)$$

$$\mathcal{I}_B(q,k,R) = \beta_0\int_R^{\infty} j_0(kr)[j_0(kr)-\tan(\delta_0(k))y_0(kr)] \qquad (37)$$

Note that the normalization constants α_0 and β_0, and the integrals $\mathcal{I}_A(q,k,R)$ and $\mathcal{I}_B(q,k,R)$ can be calculated numerically. For $\mathcal{I}_B$, $\tan\delta_0(k)$ is given by the following.

$$\tan\delta_0(k) = \frac{k\tan(qR)-q\tan(kR)}{q+k\tan(kR)\tan(qR)} \qquad (38)$$

Also note that in evaluating the above quantities, we must use values of k and q such that,

$$k = k_{\text{res}} \approx \sqrt{\frac{2mE_{\text{res}}}{\hbar^2}} \qquad (39)$$

$$q = q_{\text{res}} \approx \sqrt{\frac{2m(E_{\text{res}}+U)}{\hbar^2}} \qquad (40)$$

This is because the free states $\psi_k(E_{\text{res}})$ are evaluated at $E = E_{\text{res}}$ We will now calculate the density of states $\mathcal{D}(E_{\text{res}})$.

$$\begin{aligned}
\mathcal{D}(E) &= \int d^3k\delta(E-E_k) \qquad (41) \\
&= \int d\Omega\int k^2dk\delta\left(E-\frac{\hbar^2k^2}{2m}\right) \qquad (42) \\
&= (4\pi)\left(\frac{2m}{\hbar^2}\right)\sqrt{\frac{m}{2\hbar^2}}\int\sqrt{E'}\delta(E-E')dE' \qquad (43) \\
&= 2\pi\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E} \qquad (44)
\end{aligned}$$

$$\therefore \mathcal{D}(E_{\text{res}}) = 2\pi\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E_{\text{res}}} \qquad (45)$$

We can now put everything together by plugging equation (35) and (45) into (1).

$$\begin{aligned}
\kappa &= \frac{2\pi}{\hbar}|\langle\psi_k(E_{\text{res}})|\hat{V}_1|\varphi\rangle|^2\mathcal{D}(E_{\text{res}}) \\
&= \frac{2\pi}{\hbar}\left(\frac{l(l+1)\hbar^2}{2\pi m\sqrt{2}}\left[\mathcal{I}_A(q_{\text{res}},k_{\text{res}},R)\right.\right. \\
&\quad \left.\left.+\mathcal{I}_B(q_{\text{res}},k_{\text{res}},R)\right]\right)^2 2\pi\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E_{\text{res}}} \\
&= \frac{2(2m)^{3/2}l^2(l+1)^2}{4\hbar m^2}\sqrt{E_{\text{res}}}\left[\mathcal{I}_A(q_{\text{res}},k_{\text{res}},R)\right. \\
&\quad \left.+\mathcal{I}_B(q_{\text{res}},k_{\text{res}},R)\right]^2
\end{aligned}$$

In our code, we set $\hbar = m = 1$. The above equation would therefore reduce to the following.

$$\begin{aligned}
\kappa = 2^{1/2}[l^2(l+1)^2]\sqrt{E_{\text{res}}}\left[\mathcal{I}_A(q_{\text{res}},k_{\text{res}},R)\right. \\
\left.+\mathcal{I}_B(q_{\text{res}},k_{\text{res}},R)\right]^2
\end{aligned} \qquad (46)$$

2.3.2 Implementation

The implementation for the Fermi's golden rule calculation of κ is simply to numerically calculate it using equation (45). The procedure is as follows.

1. First, from the partial wave analysis calculation part, E_{res} would be obtained from the plot of σ_l as shown in figure 4.
2. k_{res} and q_{res} would then be calculated using the E_{res} value.
3. k_{res} and q_{res} would then be passed as arguments into functions that numerically calculate the integrals $\mathcal{I}_A$ and $\mathcal{I}_B$ using the definitions in equation (35) and (36).
4. Finally, κ would be calculated by substituting the required values into equation (46).

The above procedure was repeated for each scattering resonance and for each value of the parameter (i.e. R, U or l) being varied.

3 Results and Discussion

Note that in the following sections, the percentage error is defined as,

$$\text{Percentage Error} = \frac{|\kappa_{fgr}-\kappa_{pwa}|}{\kappa_{\text{pwa}}}\times 100\% \qquad (47)$$

where κ_{fgr} and κ_{pwa} are decay rates obtained via Fermi's golden rule and partial wave analysis respectively.

3.1 Angular Quantum Number l Dependence

Figure 5 shows the decay rate derived from partial wave analysis and Fermi's golden rule plotted against the angular quantum number l.

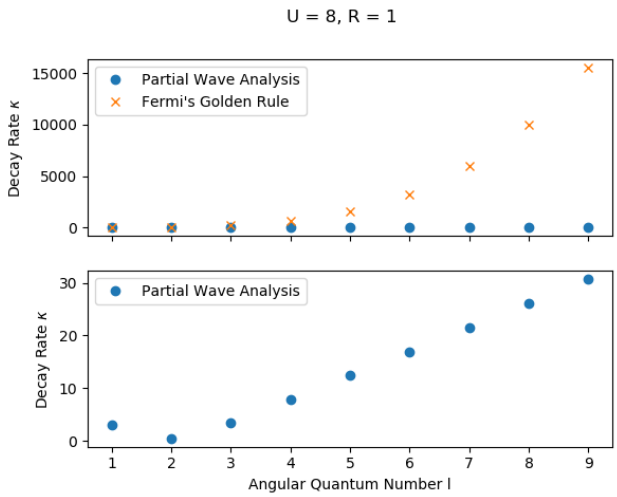

Figure 5: Decay Rate κ against l

We can see that the decay rate increases for larger values of l. This is because the well becomes narrower for larger values of l as shown in figure 6, and the particle decays faster in a narrower well.

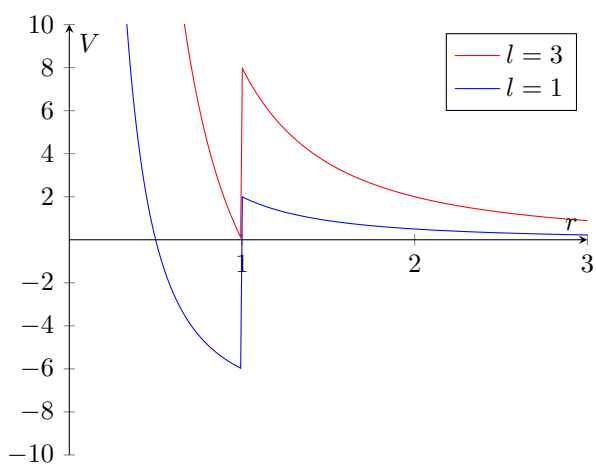

Figure 6: Effective potential $V(r)$ of a spherical well

The decay rate calculated with Fermi's golden rule also becomes less accurate as l increases. Fermi's golden rule is more accurate for a wider well whereby the quasi-bound state is more well defined. Our approximation of the free states being plane waves will also affect the accuracy of Fermi's golden rule.

3.2 Potential U Dependence

Next, we varied the potential U at $R = 1$ and $l = 1$. We set $l = 1$ as Fermi's golden rule is more accurate at small l values, based on figure 5.

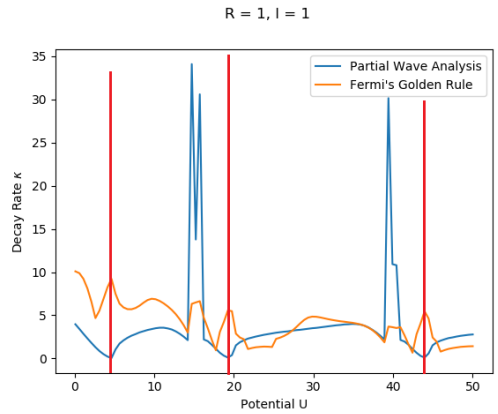

Figure 7: Decay Rate κ against U

From figure 7, it can be seen that the Fermi's golden rule approximation of the decay rate is more accurate at larger values of U, albeit only to a small extent. This might be because the well width increases slightly for larger values of U as shown in figure 8. Since the quasi-bound states are more well defined for larger U, Fermi's golden rule calculations are more accurate.

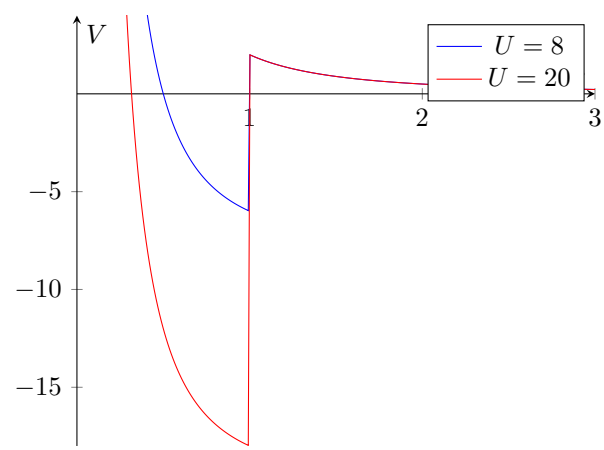

Figure 8: Effective potential $V(r)$ of a spherical well

The red lines in figure 7 shows the potential with the largest percentage error as shown in figure 9. If we plot the peak of the partial cross sections σ_l against U, we find that the spikes in the percentage error coincides with the spikes in the peak of σ_l as shown in figure 10. These spikes in σ_l signifies a strong resonance effect and thus a low decay rate. The Fermi's golden rule approximations however, have the largest percentage error at these small decay rates. In figure 10, the orange plots are calculated from equation (29). Whereas the blue plots are estimated from the σ_l calculated with partial wave analysis. As it is not obvious from figure 9, we would like to note that the percentage error is decreasing for increasing U as shown in Figure 11.

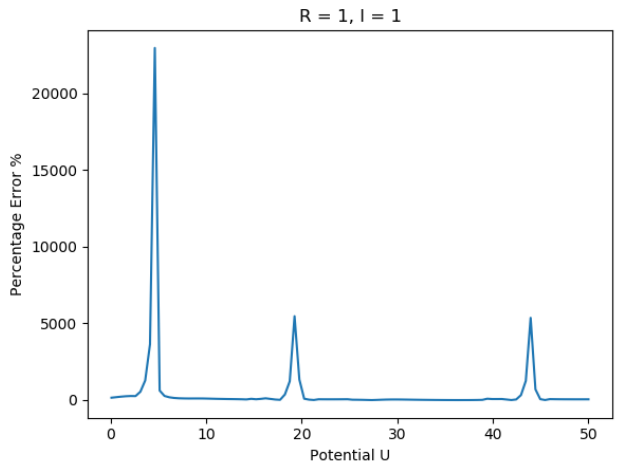

Figure 9: Percentage error of decay rate against U

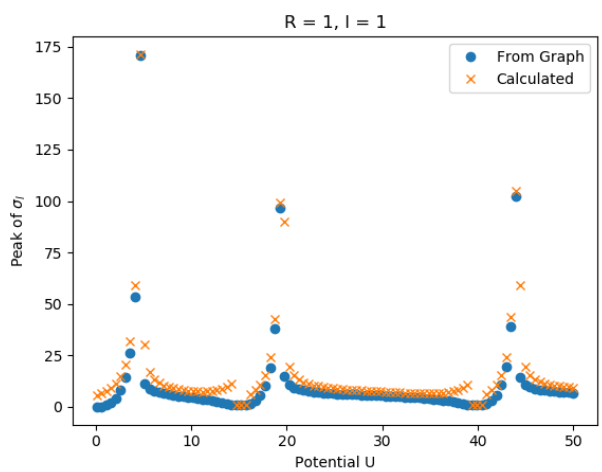

Figure 10: σ_l peaks against U

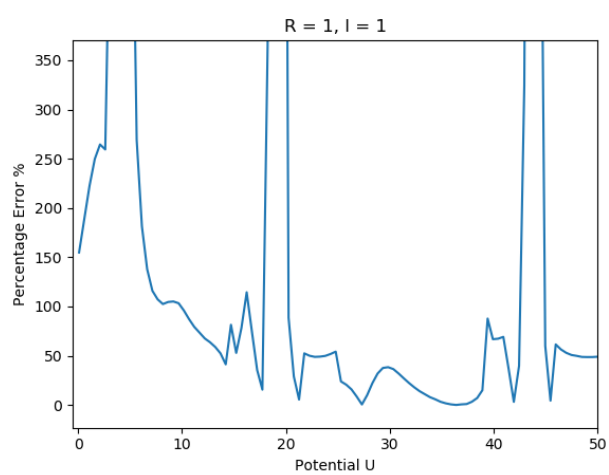

Figure 11: Percentage error of decay rate against U (Zoomed)

3.3 Radius R Dependence

Similarly for radius R dependence, we would expect the decay rate to be smaller for a wider well and thus a larger R as shown in figure 12. Again l is set to 1 and U is set to a large potential to minimize the error of the Fermi's golden rule approximation. The error of the Fermi's golden rule approximation is increasing for increasing R as shown in figure 13. This is because even though the well is getting wider, the height is getting smaller as shown in figure 14, hence the quasi-bound states are not well defined for large values of R and disappears completely as $R \to \infty$. This is different from increasing U shown in figure 8, whereby increasing U only increases the well's width and does not decreases the height of the well.

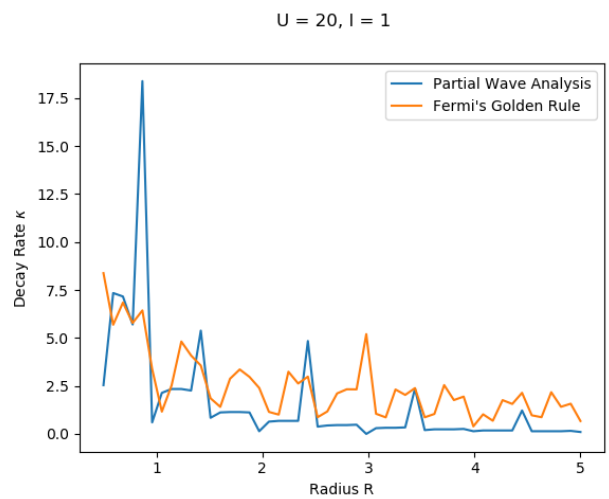

Figure 12: Decay rate κ against R

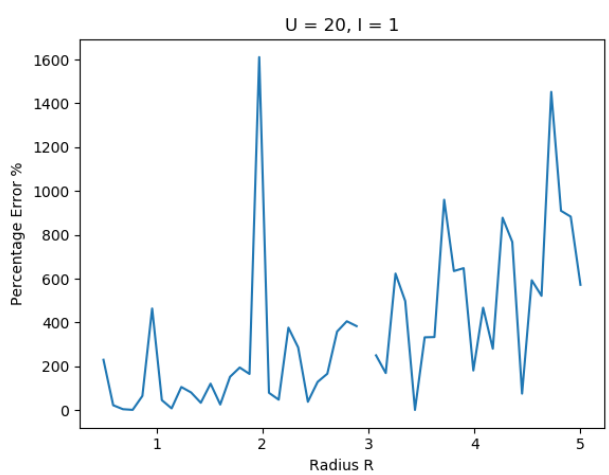

Figure 13: Percentage error of decay rate against R

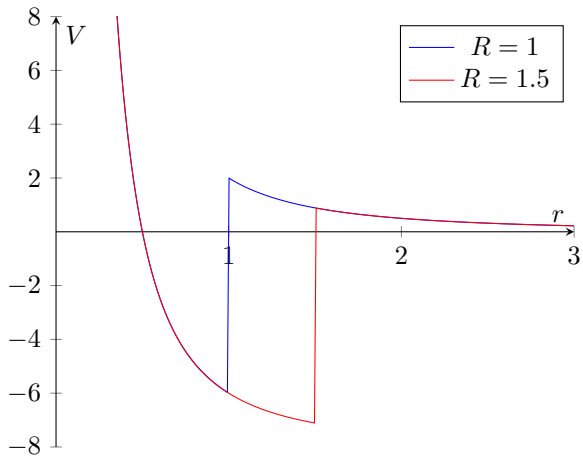

Figure 14: Effective potential $V(r)$ of a spherical well

From the results we can conclude that Fermi's golden rule is more accurate for a well defined well with quasi-bound states. To be well defined, the well cannot be too narrow, hence l cannot be too large. The height of the well must also be high enough for the quasi-bound states to be well defined, thus R must also not be too large. Lastly, to increase the width of the well, a larger potential U will result in a larger width. We approximated $l \approx 1$, $R \approx 1$ and a large $U \geq 20$ for Fermi's golden rule to be accurate.

4 Conclusion

In conclusion, the decay times for the quasi-bound states of a spherically symmetric attractive well were approximated using Fermi's golden rule. These decay times were then compared against decay times calculated using partial wave analysis in order to evaluate the accuracy of Fermi's golden rule. From the results, it was concluded that Fermi's golden rule is reasonably accurate for a well with parameters $l \approx 1$, $R \approx 1$ and $U \geq 20$. Qualitatively, Fermi's golden rule is accurate for wells which support a large range of energies which correspond to quasi-bound states. These energies are energies in which a particle can escape the well via quantum tunneling. Our results also show that a well with a small widths resulted in the quasi-bound states decaying much faster. One limitation of our study was that the free states $\psi_k(E_{\text{res}})$ were approximated as plane waves travelling in the z-direction when calculating the transition amplitude for Fermi's golden rule. A possible improvement would be to approximate $\psi_k(E_{\text{res}})$ as outgoing spherical waves. Future work would be a more detailed investigation on how the width of the spherically symmetric well affects the accuracy of Fermi's golden rule. Other spherically symmetric wells such as the dirac-shell potential can also be studied using the same methodology as this study.

References

Bransden BH, Joachain CJ. Quantum mechanics. Uttar Pradesh, India: Pearson; 2017.

Appendix

```python
import matplotlib.pyplot as plt
import scipy as sp
from scipy.special import spherical_jn, spherical_kn, spherical_yn
from scipy.integrate import quad


def scattering_amplitude(Evec, U, R, ll):
    """Returns sigma for a range of energies"""

    fl = sp.zeros(len(Evec), dtype=complex)
    sigma = sp.zeros(len(Evec), dtype=complex)

    kR = sp.sqrt(2 * Evec) * R
    qR = sp.sqrt(2 * (Evec + U)) * R

    # Spherical bessels/hankels, and their derivatives
    jl = spherical_jn(ll, qR)
    jlp = spherical_jn(ll, qR, derivative=True)
    hl = spherical_jn(ll, kR) + 1j * spherical_yn(ll, kR)
    hlp = spherical_jn(ll, kR, True) + 1j * spherical_yn(ll, kR, True)

    delt = 0.5 * sp.pi - sp.angle((kR * hlp * jl - qR * hl * jlp) / R)
    fl = (-0.5j * R / kR) * (sp.exp(2j * delt) - 1) * (2 * ll + 1)
    sigma = abs(fl)**2 * 4 * sp.pi / (2 * ll + 1)

    return sigma


def FWHM(X, Y):
    """Returns the FWHM, halfmax and max X"""

    d = Y - (max(Y) / 2)
    indexes = sp.where(d > 0)[0]
    max_index = sp.argmax(Y)

    return X[indexes[-1]], X[indexes[0]], max(Y) / 2, X[max_index]


def fermi_calc(Eres, qres, kres, Rvec, lvec, var):
    """Returns the decay rate approximated by Fermi's Golden Rule."""

    def integral_A(q, k, R, ll):
        # Find the normalization constant first
        integrand0 = lambda r: spherical_jn(ll, q * r)**2
        alpha, err = quad(integrand0, 0, R)
        alpha = pow(alpha, -0.5)

        # Define integrand of integral A
        integrand1 = lambda r: alpha*spherical_jn(ll, k*r)*spherical_jn(ll, q*r)

        Ia, err = quad(integrand1, 0, R)

        return Ia

    def integral_B(q, k, R, ll):
        # Find the normalization constant first
        tan_delt = (k * sp.tan(q * R) - q * sp.tan(k * R)) / (
            q + k * sp.tan(k * R) * sp.tan(q * R))
        integrand0 = lambda r : (spherical_jn(ll, k*r)-tan_delt*spherical_yn(
            ll, k*r))**2
        beta, err = quad(integrand0, R, sp.inf, limit=8000)
        beta = pow(beta, -0.5)

        # define integrand of integral B
        integrand1 = lambda r: beta*spherical_jn(ll, k*r)*(
            spherical_jn(ll, k*r)-tan_delt*spherical_yn(ll, k*r))

        Ib, err = quad(integrand1, R, sp.inf, limit=8000)

```

```
        return Ib

    kappa = sp.zeros(len(Eres))
    for i in range(len(Eres)):
        if var == 'R':
            R = Rvec[i]
            l = lvec
        elif var == 'l':
            R = Rvec
            l = lvec[i]
        else:
            R = Rvec
            l = lvec
        Ia = integral_A(qres[i], kres[i], R, 0)
        Ib = integral_B(qres[i], kres[i], R, 0)

        kappa[i] = (2**(5 / 2)) * (l**2) * ((l + 1)**2) * sp.sqrt(Eres[i]) * ((
            Ia + Ib)**2) / 4

    return kappa


def main(Uvec, Rvec, lvec, var='U', Emax=100):
    """Plot and compare decay rates against a given variable.

    Set var to the variable to plot against -> 'U', 'R', 'l'.
    """

    Evec = sp.linspace(1e-6, Emax, 100000)
    sigma = []
    Uvec_len = sp.array(Uvec).size
    Rvec_len = sp.array(Rvec).size
    lvec_len = sp.array(lvec).size

    # Find sigma for different variables
    if var == 'U' and Uvec_len > 1 and Rvec_len == 1 and lvec_len == 1:
        for U in Uvec:
            sigma.append(scattering_amplitude(Evec, U, Rvec, lvec))
            xlabel = "Potential U"
            title = "R = {}, l = {}".format(Rvec, lvec)
            vec = Uvec

    elif var == 'R' and Rvec_len > 1 and Uvec_len == 1 and lvec_len == 1:
        for R in Rvec:
            sigma.append(scattering_amplitude(Evec, Uvec, R, lvec))
            xlabel = "Radius R"
            title = "U = {}, l = {}".format(Uvec, lvec)
            vec = Rvec

    elif var == 'l' and lvec_len > 1 and Uvec_len == 1 and Rvec_len == 1:
        for l in lvec:
            sigma.append(scattering_amplitude(Evec, Uvec, Rvec, l))
            xlabel = "Angular Quantum Number l"
            title = "U = {}, R = {}".format(Uvec, Rvec)
            vec = lvec

    else:
        print("Invalid entries.\n"
              "Either var is not set to 'U', 'R' or 'l',"
              "\nor more than 1 variable is a vector"
              "\nor the chosen variable is not a vector.")
        return

    sigma = sp.array(sigma)

    # Find FWHM, Eres, kres and qres
    Eres = sp.zeros(len(sigma))
    FW1 = sp.zeros(len(sigma))
    FW2 = sp.zeros(len(sigma))
    for i in range(len(sigma)):
        FW1[i] = FWHM(Evec, sigma[i])[0]
```

```python
        FW2[i] = FWHM(Evec, sigma[i])[1]
        Eres[i] = FWHM(Evec, sigma[i])[3]
    FWHM_data = (Eres - FW2) * 2
    kres = sp.sqrt(2 * Eres)
    qres = sp.sqrt(2 * (Eres + Uvec))

    # Calculate decay rate with Fermi's Golden Rule
    kappa_fermi = sp.zeros(len(sigma))
    kappa_fermi = fermi_calc(Eres, qres, kres, Rvec, lvec, var)

    # Calculate percentage error
    per_err = abs(FWHM_data - kappa_fermi) / FWHM_data * 100

    # Plot decay rate comparison
    plt.figure()
    plt.plot(vec, FWHM_data, '-', label='Partial Wave Analysis')
    plt.plot(vec, kappa_fermi, '-', label='Fermi\'s Golden Rule')
    plt.ylabel("Decay Rate $\kappa$")
    plt.xlabel(xlabel)
    plt.legend()
    plt.title(title)

    # Plot percentage error
    plt.figure()
    plt.plot(vec, per_err, '-')
    plt.ylabel("Percentage Error %")
    plt.xlabel(xlabel)
    plt.title(title)

    # Plot peak comparison
    peak = sp.zeros(len(sigma))
    peak_cal = sp.zeros(len(sigma))
    for i in range(0, len(sigma)):
        peak[i] = max(sigma[i])
        if var == 'l':
            peak_cal[i] = 4 * sp.pi * (2 * lvec[i] + 1) / (kres[i])**2
        else:
            peak_cal[i] = 4 * sp.pi * (2 * lvec + 1) / (kres[i])**2

    plt.figure()
    plt.plot(vec, peak, 'o', label="From Graph")
    plt.plot(vec, peak_cal, 'x', label="Calculated")
    plt.ylabel("Peak of $\sigma_l$")
    plt.xlabel(xlabel)
    plt.legend()
    plt.title(title)

    # Plot a single sigma with FWHM
    index = 1
    half_max = max(sigma[index]) / 2
    plt.figure()
    plt.plot(Evec, sigma[index])
    plt.plot([FW1[index], FW2[index]], [half_max, half_max], label='FWHM')
    plt.plot(
        [Eres[index], Eres[index]], [0, max(sigma[index])], label='$E_{res}$')
    plt.ylabel("Partial Cross Section $\sigma_l$")
    plt.xlabel("Energy E")
    plt.legend()
    plt.title(title)


# Parameters
# Only 1 is allowed to be a vector
Uvec = 20
Rvec = sp.linspace(0.5, 5, 50)
lvec = 1

# Set var = 'U', 'R' or 'l'
main(Uvec, Rvec, lvec, var='R', Emax=1000)
plt.show()
```